Journal of Computer-Aided Molecular Design (2026) 40:84
https://doi.org/10.1007/s10822-026-00792-8

Predicting drug-target interactions and binding affinity using an optimized deep learning approach

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Received: 24 November 2025 / Accepted: 12 March 2026
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Abstract

The accurate prediction of Drug-Target Interactions (DTIs) and Drug-Target Affinity (DTA) is crucial for reducing experimental costs and time, thereby accelerating drug discovery and repurposing efforts. The local biochemical contexts and the global structural dependencies between drugs and their target proteins are often insufficiently captured by conventional deep learning models, which restrict their predictive performance. In this work, we introduce a CNN based Dual Attention (nCNN-DA), which combines 1D convolutional feature extraction with channel and spatial attention mechanisms to enhance the representational power of features for drug SMILES and protein sequences. The model was trained and tested on three benchmark datasets: KIBA, Davis, and BindingDB, using AUPR, AUROC, MSE, Pearson correlation, and accuracy. Experimental results show that nCNN-DA significantly improves performance compared to well-established models (FusionNet, GraphormerDTI, DeepDTAGen, and DTBA-net). In particular, nCNN-DA achieved the best accuracy of 98.5%, 95.5%, and 97.5%, as well as the lowest MSE of 0.1559, 0.3189, and 0.2957 on KIBA, Davis, and BindingDB, respectively, and better scores for AUPR and Pearson Correlation. These findings further demonstrate that nCNN-DA issued for identifying putative DTI pairs and predicting binding affinities with high quality, making it a versatile and general method for drug discovery, virtual screening, and drug repurposing.

Keywords Drug target interaction · Drug target affinity · Dual attention mechanism · nCNN-DA · Deep learning approach

Introduction

Drug discovery is the procedure to identify new drugs for treating any disease. Historically, drug discovery is complex, time consuming and resource intensive [1]. Notably, for the success of this campaign, it is essential to predict drug-target interactions (DTIs) and binding affinities (DTAs), which significantly support both the efficacy and safety of potential drugs [2]. Solid methods, such as X-ray crystallography, NMR spectroscopy, and isothermal titration calorimetry, provide high-precision experimental descriptions of the interactions between proteins and ligands. However, these methods are time-consuming and costly, making them unsuitable for high-throughput screening. As a result, computational tools for predicting DTIs and DTAs have emerged as essential tools in facilitating drug discovery, repositioning, and precision medicine [3–4].

The conventional machine learning models, such as SimBoost and KronRLS, have achieved moderate performance in DTI prediction using hand-crafted molecular descriptors and similarity matrices [5]. However, the relationship between drug molecules and protein targets is weak and non-linear at both the structural and biochemical levels [6]. Deep learning techniques, including DeepDTA and GraphDTA, have been proposed to directly learn feature representations from raw SMILES strings and amino acid sequences, thereby improving predictive

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Published online: 23 March 2026

performance. However, despite the improvement, three main limitations remain in the existing techniques [7–8]:

- Extracting limited structural information, 1D-CNN and other sequence methods are unable to incorporate complete 3D structures and long-range interactions.
- Inadequate modelling of the global context, models lack an attention mechanism to highlight important atom-residue pairs while ignoring irrelevant ones.
- The lack of consistent performance across datasets means models are suffering from dataset-specific overfitting and cannot generalize well for unseen drug-target pairs.

Traditional methods for DTA prediction such as similarity-based machine learning, and regression models including SimBoost or linear scoring functions that were developed using hand-crafted molecular descriptors [9]. With the rise of deep learning, CNN based models (DeepDTA), graph neural network methods and transformer-based architectures like TransVAE-DTA were proposed to capture further structural and long-range interaction features [10]. Meanwhile, network-based and AI-enhanced models for DTI/DTA and drug repurposing have shown the increased performance in learning representation and predictive robustness by using techniques such as large-scale AI model [11–12]. These limitations also indicate a requirement for a powerful deep learning paradigm that can improve not only local feature learning but also global interaction modeling, while remaining robust across multiple datasets. Against this background, we propose our nCNN-DA model, which combines a dual attention mechanism with a well-designed CNN feature extractor to model fine-grained molecular patterns and capture long-range dependencies, aiming to offer a more precise, interpretable, and generalizable solution for DTI and DTA prediction [13–14].

The objective of nCNN-DA is to enhance predictive accuracy for DTI and DTA takes, the integration of the dual attention mechanism (Channel Attention and Spatial Attention) also enables qualitative interpretability of learned interaction patterns. To support this, we visualized spatial attention weights over drug SMILES tokens and protein amino acid sequences for representative drug-targets pairs. The model presented attempts to overcome the limitations and shortcomings of previous methods, which fail to properly represent features, do not generalize well to unseen drug–target pairs, and exhibit high prediction errors.

The main contributions of this research are;

- Presented a nCNN-DA model, the convolutional neural network combining with the Dual Attention mechanism (channel and spatial attention) for the accurate prediction of DTI and DTA.
- The convolutional network, having been optimized, takes into account local biochemical signals, whereas the dual attention mechanism aggregates important structural and functional interactions between drug molecules and target proteins.
- Comparisons to the state-of-the-art models (DTBA-net, DeepDTAGen, GraphormerDTI, and FusionNet) under the measures of MSE, AUPR, AUROC, Pearson correlation, and accuracy.
- Tested on three benchmark datasets such as BindingDB, KIBA, and Davis, achieving good generalization and robustness across different datasets.

The rest of this paper is structured as follows: Sect. "Background study" represents the background study. Section "Materials and methods" explains the materials and methods. Section "Experimental setup and simulation parameters" represents the experimental setup, including performance measurements and outcomes. Finally, Sect. "Conclusion" presents the conclusion and outlines the future scope of this work.

Background study

The prediction of drug–target interactions (DTIs) and drug–target affinities (DTAs) is recognized as an essential step in drug discovery and repositioning. In recent years, various machine learning (ML) and deep learning (DL) methods have been introduced to enhance the accuracy and scalability of DTI prediction models. Deep learning has revolutionized this field, where complex features can be learnt directly from the raw inputs, e.g., SMILES and amino acid sequences [15–16]. CNNs do learn sequence-level representations, performing better than traditional ML methods. Yet, these techniques generally fail to aggregate global structure information and long-range dependencies in the sequence of molecules and proteins [17–18].

Mishra et al. [19] proposed the DTBA-net, a hybrid deep learning model that utilizes a feature selection technique in conjunction with a CNN for the efficient prediction of Drug-Target Binding Affinity (DTBA). The model also integrates the Modified JAYA Algorithm, utilizing protein sequences and compound structures to select the best features, which helps reduce computational power and address the problem of overfitting. Comprehensive evaluations on benchmark data sets, including KIBA and Davis, demonstrated the robustness of the model, which outperformed state-of-the-art methods with an R^2 value of greater than 95% and a Mean Absolute Error (MAE) of 0.17. The study highlights the importance of integrating feature selection strategies with deep learning models to enhance prediction accuracy.

Shah et al. [20] proposed DeepDTAGen, a multitask deep learning network that can simultaneously predict Drug–Target affinity (DTA) and generate drug-aware targets. The model shares the feature space for both tasks and learns the underlying ligand–receptor interplay. A fundamental technique of DeepDTAGen is FetterGrad, which addresses the optimization difficulties caused by gradient interference among tasks. Through validations on KIBA, Davis, and BindingDB, DeepDTAGen outperformed other state-of-the-art methods, with MSEs of 0.146, 0.214, and 0.458, respectively. The multitask learning approach demonstrated better generalization and stability; however, the learning cost of these models was high, and optimization remained an issue.

Gao et al. [21] presented GraphormerDTI—a graph transformer method that can improve predictions of drug-target interactions (DTIs) by handling the structure of molecular graphs properly. The approach incorporates node centrality encoding, spatial encoding, and edge encoding to learn important structural information, such as molecular subgraph patterns and chemical bond semantics. In the protein representation, 1D-CNN with attention mechanism was used to enable efficient drug–protein interaction modeling. Experiments on three benchmark datasets demonstrated that the proposed GraphormerDTI achieved more accurate predictions than several state-of-the-art baselines, particularly for out-of-sample molecular predictions, thereby making it applicable to virtual screening in drug development. Nevertheless, the large computational complexity of graph transformer layers, as well as the necessity of extensive label annotations, has become a significant bottleneck to the practical application of triplet-based learning.

Vijayan et al. [22] proposed FusionNet, a hybrid deep learning model including CNN, LSTM and Transformer for enhanced DTBA prediction, a hybrid deep learning including CNN, LSTM and Transformer model for enhanced DTBA prediction CNNs for local sequence-level feature extraction, LSTMs for the sequential dependencies, and Transformers for the long-range contextual interaction between drugs and protein. Moreover, the training stability and convergence, especially under the large-batch regime, were further improved by using the Layer-wise Adaptive Moments (LAMB) optimizer. FusionNet produced an MSE of 0.20 for Davis and 0.18 for KIBA, and surpassing previous models such as SimBoost and DeepDTA.

Wang et al. [23] introduced DeepDTAF, a deep learning-based approach to predicting protein–ligand binding affinity. The model combines sequence-based feature extraction with deep convolutional layers to capture the spatial as well as the sequential relationship between the ligands and protein sequences. Through this end-to-end training, DeepDTAF also significantly outperformed classical machine learning approaches for binding affinity prediction. However, the presence of only sequence representations of the drug impedes the capturing of information about the graph structures of molecules, which are fundamental for understanding the topological relationships between atoms and with other similar charged molecules.

Wang et al. [24] designed a well-tuned graph neural network (GNN), GraphscoreDTA, for protein-ligand binding affinity prediction. The drug molecules are modeled as hippocampal networks, treating them as graphs and leveraging node-level and graph-level message-passing to capture structural subgraph patterns. Sequence-based encoders encode the protein sequences. GraphscoreDTA outperformed existing GNN-based models by introducing optimized scoring strategies for the process of structural feature aggregation. However, due to its particularly high computational complexity and much graph construction and pre-processing, the approach may not scale well for large datasets.

Sun et al. [25] proposed AGCN-based drug repositioning approach. This approach builds a heterogeneous drug–target graph that adaptively adjusts graph convolution weights by capturing the most informative drug–target associations. The method achieves competitive prediction capabilities on the prediction of new drug–target interactions to reposition drugs, showing the efficiency of

Table 1 Reference study of advantages and limitations on various models with datasets

References	Dataset used	Method	Objective	Advantages	Limitations
Lu et al. [27]	Yamanishi_08, Hetionet, Davis, KIBA,	DTIAM Self-supervised multitask learning with pre-training	Predict DTI, DTA, and mechanisms of action (MoA)	Strong generalization; effective cold-start performance; MoA differentiation	Requires large pre-training data; computationally expensive
Zeng et al. [28]	Multiple curated DTI datasets	Drug-Online GNN and CNN platform	Online prediction of DTI, DTA, and binding sites	Combines interaction, affinity, and binding site prediction; useful for drug design	High dependency on curated datasets; not optimized for high-throughput screening
Bhatia et al. [29]	KIBA, Davis	CSAN-BiLSTM-Att optimized with Differential Evolution	Predict drug target binding affinities for drug repositioning	High performance (CI 0.971, MSE 0.014 on KIBA); robust hyperparameter tuning	Complex architecture; high training cost
Ranjan et al. [30]	KIBA, Davis	MDF-DTA Multi-Dimensional Fusion (1D, 2D, 3D representations)	Integrate multi-dimensional drug and target features for DTA prediction	Outperforms state-of-the-art models; feature importance demonstrated by ablation studies	Increased computational complexity; relies on pre-trained models
Pragya et al. [31]	BindingDB, Davis, PDAC-specific datasets	RFR, XGBR, 1D-CNN with molecular descriptors	Disease-specific binding affinity prediction and drug repurposing for PDAC	Effective for disease-specific targets; interpretable descriptor-based approach	Less generalizable to other diseases; higher MSE on PDAC dataset
D'Souza et al. [32]	Public chemogenomics datasets	DeepPS, Deep CNN using protein binding site information	Predict DTIs using SMILES and motif-rich protein pocket subsequences	Incorporates binding site information; computationally efficient	Limited to pocket-level features; may miss long-range dependencies
Wu et al. [33]	DrugBank, AD-specific datasets	HNNDTA, Hybrid Neural Network with molecular docking	Predict DTA for drug repurposing in Alzheimer Disease	Combined DTA prediction and docking; identified potential AD leads	Domain-specific; requires docking validation; moderate scalability

Table 2 Comparison study of drug and target representation on various methods

Study	Method category	Drug representation	Target representation	Key innovation	Strength	Limitation
Mayr et al. [34]	Large-scale ML comparison	Molecular descriptors (ChEMBL)	Assay-based features	Nested cluster cross-validation to reduce compound bias	Robust benchmarking; DL outperformed classical ML	No structural modeling; limited interpretability
He et al. [35]	Hybrid sequence–graph DTA	SMILES tokens + molecular graph	Protein sequence + contact graph	Cross-modality feature interaction via graph modeling	Combines sequence & structure information	Higher computational complexity
Lu et al. [36]	Unified DTI/DTA/MoA framework	Molecular graph + self-supervised embeddings	Protein sequence (Transformer pretraining)	Multi-task self-supervised pretraining	Strong generalization; handles cold-start	High training cost; complex architecture
He et al. [37]	Dual-modality DTA	SMILES + drug graph	Binding-site-focused protein graph (AF2-based)	Binding-site-centric graph balancing	Improved unseen drug/target performance	Depends on structure/contact prediction accuracy
Gupta et al. [38]	Network-based drug repurposing	Drug-target network topology	Single-cell gene regulatory networks	Cell-type-specific GRN-based modeling	Precision medicine potential	Indirect affinity modeling
He et al. [39]	Interpretable AI for drug repurposing	Gene expression signatures	Cell-type-specific gene activation curves	Kolmogorov–Arnold Network (KAN) for interpretability	High interpretability; translational relevance	Not specifically designed for DTA regression

the graph-based learning on the DTI tasks. But AGCNs mainly concentrate on interaction prediction rather than quantitative binding affinity estimation, and thus may not be as applicable for precise affinity prediction tasks.

Li et al. [26] introduced BACPI related to CPPI (Bi-directional Attention Neural Network for Compound–Protein Interaction and Binding Affinity Prediction), which constructs bidirectional attentive mechanisms to capture the mutual interaction information between compound atoms and protein residues. Through adopting context-aware interactions, BACPI can be more interpretable and can provide better derived predictive performance against the

Table 3 Dataset details

Dataset	No. of drug molecules	No. of protein targets	No. of drug target interaction pairs	Affinity type	Measurement unit	Processed format
KIBA	2116	229	118,254	KIBA Score (integrated from Ki, Kd, IC50)	Arbitrary KIBA Score	Normalized continuous values
Davis	68	442	30,056	Dissociation Constant (Kd)	nM (converted to pKd)	pKd = $-\log10(Kd/10^9)$
BindingDB	5200	1700	129,000	Ki, Kd, IC50	nM (converted to pKi or pKd)	Normalized continuous values

classical CNN-based models. However, using only attention mechanisms is lead to sub-optimal structural feature extraction in the case of complex molecules. Table 1 shows the reference study of advantages and limitations on various models with datasets and Table 2 shows the comparison study of drug and target representation on various methods.

The nCNN-DA model attempts to address these drawbacks by incorporating dual attention (channel and spatial attention) within a well-designed CNN structure. This design affords the model the capability to co-learn fine-grained local biochemical information and global structural relationships, leading to increased predictive performance, interpretability and generalization across a variety of benchmark datasets.

Materials and methods

Datasets

The proposed nCNN-DA model was assessed by three commonly-used benchmark datasets for drug–target interaction and binding affinity prediction (KIBA, Davis and BindingDB) [40–43]. These datasets were chosen because they had large coverage of drug–target pairs and the validated experimental affinity values lead to a trustworthy model training and testing.

The KIBA dataset is curated for kinase inhibitor activity studies, comprising different bioactivity data (Ki, Kd, and IC50) and converted to a consistent score (KIBA score). It includes 118,254 drug-target interactions comprising 2116 distinct drugs (kinase inhibitors) and 229 protein kinases. The KIBA dataset is especially valuable to compare the performances of regression-based models, since it presents a wide spectrum of affinity scores, enabling us to evaluate both high and low-binding interactions. The Davis dataset is specific to kinase inhibitor-target interactions, and it is generated with experimentally determined Kd (dissociation constant of the molecule) values expressed in nM scale. The dataset contains 30,056 pairs of drugs–target interactions with 68 unique kinase inhibitors and 442 protein targets. To achieve better numerical stability, which also follows a standard practice in affinity prediction tasks, we converted the Kd to pKd ($pKd = -\log10(Kd/10^9)$) values, which resulted in 7.5 mM being changed to 7.5 mM. The Davis dataset is well-known for its high-quality and clean experimental data, making it an appropriate choice for accurate affinity estimation. The BindingDB is a large collection of measured protein–ligand binding affinities across varied target classes, such as enzymes, G-protein-coupled receptors, and ion channels. In this study, a subset of the BindingDB dataset was selected, including data for which the experimental values of Ki, Kd, or IC50 are available. After data pre-processing and curation of duplicate data, the final compiled dataset included nearly 1,29,000 drug–target interaction pairs, which corresponded to over 5200 distinct drug molecules and 1700 target proteins. The incorporation of BindingDB enables model testing for generalization over diverse molecular scaffolds and target families besides kinase inhibitors. The detail features of the dataset are represented in Table 3.

Pre-processing and feature representation for data

To achieve better accuracy and generalization performance, effective data pre-processing and feature representation are key features of the proposed nCNN-DA model. The protein and ligand sequence information for the three benchmark databases (KIBA, Davis, and BindingDB) were processed to harmonize the information, exclude redundancies, and change the data into a format compatible with machine learning. All three data sets were pre-processed to remove redundant drug–target interaction pairs and to rescale affinity to be in the same range (converted as pKd or pKi). The pre-processed data were split

randomly to the 80%, 10%, 10% proportions for the training, validation, and test, respectively, with passing labels by stratification to ensure a balanced presence of strong and weak interactions.

Drug representation

Drug molecules were encoded using SMILES strings, a chemical structure representation by text that can be converted to a molecular fingerprint. From each SMILES, a tokenization obtained of meaningful chemical substructures (atoms, bonds, ring notation). Dense vector embeddings were generated from the tokenized sequences by the embedding layer, allowing the model to represent both the structural and chemical features of drug molecules. As SMILES sequences comes in different lengths, the codes were 'pre-processes' by padding shorter sequences with zeroes and truncating longer ones to a length equal to the maximum sequence distribution in the training set. This allowed for a consistent input size for the convolutional layers. In addition, numerical drug features were min–max normalized to promote the flow of gradient during the training process.

Target protein representation

From these datasets, protein sequences were collected in FASTA format. Amino acid level tokenization of the protein sequences. The protein sequence was tokenized on the amino acid level, i.e., each residue was assigned a token. As in the case of drugs, an embedding layer was used to embed amino acids in a high-dimensional continuous vector space that encodes bio-chemical and evolutionary features. Positional embeddings were included to maintain sequential order information, which is essential for interpreting long-range interactions between residues that affect binding posture. Sequences were processed and padded, allowing them to have a fixed length and enabling batch processing in deep learning models. With this standardized representation, the nCNN-DA model can effectively capture local residue patterns through convolutional layers and global context dependencies through the dAM.

Proposed nCNN-DA framework

The novel Convolutional Neursal Network with Dual Attention (nCNN-DA) architecture is proposed to effectively predict both DTIs and DTAs by jointly capturing local biochemical features as well as global structural dependencies between drug molecules and protein targets. Contrary to most traditional CNN-based models that

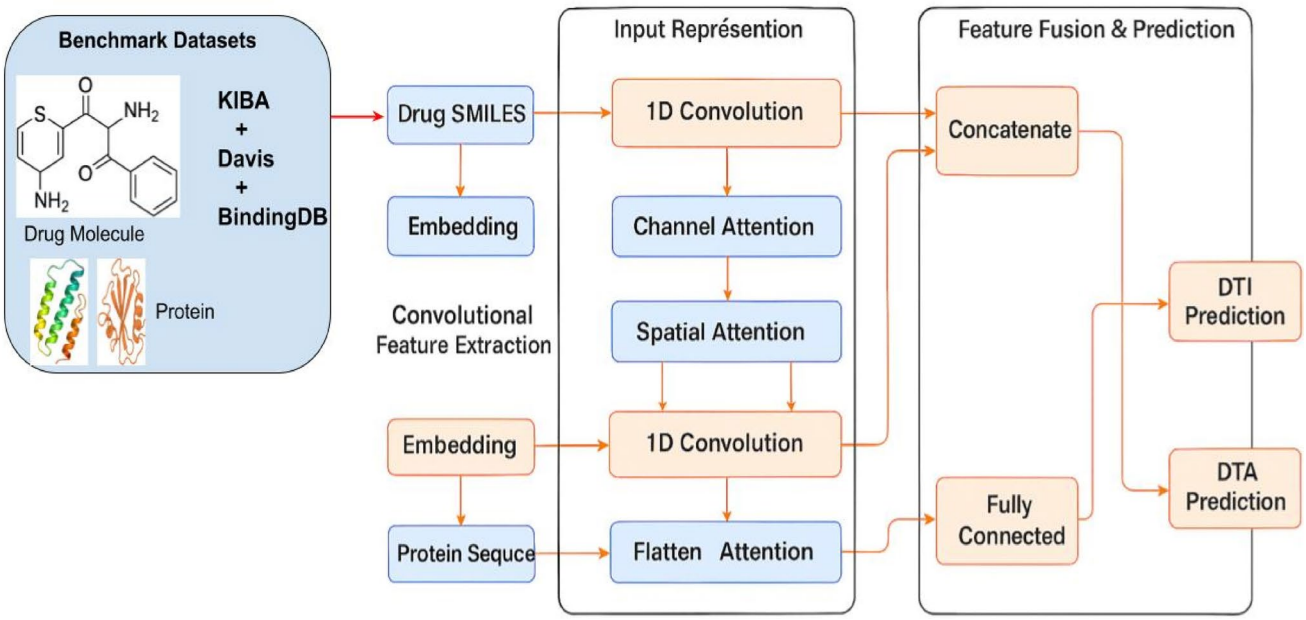


Fig. 1 Proposed cCNN-DA framework

focus on local feature extraction, the nCNN-DA incorporates a dual attention mechanism (DAM) designed to enhance feature interaction modeling, ultimately leading to improved interpretability and predictability. The Fig. 1 architecture is composed of three main parts: feature extraction with CNN layers, dual attention modules (Channel Attention & Spatial Attention), and a fusion & prediction network.

To illustrate the mathematically annotatable process by which the proposed nCNN-DA model operates, we provide a formal representation of the model, along with its specific inputs, convolution-like feature extraction, dual attention, and final outputs.

Let a drug molecule D be represented as a SMILES sequence of length L_d and a protein target P as an amino acid sequence of length L_p after tokenization and embedding using Eq. (1):

$$E_d \in \mathbb{R}^{L_d \times d_e}, \quad E_p \in \mathbb{R}^{L_p \times d_e} \tag{1}$$

where d_e is the embedding dimension, E_d and E_p are the embedded representations of drug and protein sequence, respectively.

Local features are extracted using 1D convolutional operations by Eqs. (2) and (3):

$$F_d = \sigma\ (conv1D_d\,(E_d)) \tag{2}$$

$$F_p = \sigma\ (conv1D_p\,(E_p)) \tag{3}$$

where $F_d \in \mathbb{R}^{C_d \times L'_d}$ and $F_p \in \mathbb{R}^{C_p \times L'_d}$ are feature maps, C_d and C_p are the numbers of convolutional filters, L'_d and L'_p are the sequence length after convolution, σ is the ReLU activation function.

The CA module emphasizes the important feature channels by applying global average pooling (GAP) and global max polling (GMP) followed by shared multi-layer perceptions (MLP) in Eq. (4):

$$M_c = \sigma\ (MLP\,(GAP(F)) + MLP\,(GMP\,(F)) \tag{4}$$

The refined features are obtained by:

$$F^{CA} = M_c\ \odot\ F \tag{5}$$

where $\odot$ denotes element wise multiplication, and F represents F_d or F_p.

The SA modules capture the key interaction regions by generating a spatial attention map using average and max pooling along the channel dimensions by Eqs. (6) and (7):

$$M_s = \sigma\ (conv2D\,([AvgPool(F)^{CA}, MaxPool(F)^{CA}])) \tag{6}$$

$$F^{SA} = M_s\ \odot\ F^{CA} \tag{7}$$

Thus, F_d^{SA} and F_p^{SA} represents dual attention-refined feature maps for drugs and protein sequences.

The refined protein and drug features are flattened and concatenated by Eq. (8):

$$z = [Flatten\,(F_d^{SA})\,, Flatten\,(F_p^{SA})] \tag{8}$$

This fused representation is passed through fully connected layers using Eq. (9):

$$h = \sigma\ (W_h Z + b_h) \tag{9}$$

Binary DTI prediction using Eq. (10):

$$\widehat{y}_{DTI} = Sigmoid\,(W_c h + b_c) \tag{10}$$

Binding Affinity (DTA) prediction by Eq. (11):

$$\widehat{y}_{DTA} = W_r h + b_r \tag{11}$$

A hybrid loss combining binary cross-entropy (BCE) for DTI prediction and mean squared error (MSE) for DTI prediction is used by Eq. (12):

$$\begin{aligned}\mathcal{L} = \lambda_1 * [&-y \log(\hat{y}_{DTI}) - (1-y)\log(1-\hat{y}_{DTI}) \\ &+\lambda_2 * (y_{DTI} - \hat{y}_{DTI})]\end{aligned} \tag{12}$$

where λ_1 and λ_2 are weighting coefficients, y is the true interaction label, and y_{DTI} is the experimental binding affinity value.

The pseudocode of the nCNN-DA (novel Convolutional Neural Network with Dual Attention) method for predicting the DTI and DTA is follows:

Table 4 Simulation parameters

Parameter	Value
Initial learning rate	1.00E−04
Learning rate decay	0.95 decay every 10 epochs
Batch size	64
Epochs	40
Early stopping	Stop after 10 epochs without validation improvement
Optimizer	Adam
Dropout rate	0.3 (applied to fully connected layers)
Loss function	Hybrid: BCE (DTI) and MSE (DTA)
Activation functions	ReLU, Sigmoid (DTI), Linear (DTA)
Weight initialization	Xavier Initialization

Input: D: Drug SMILE sequences, P: Protein amino acid sequences, y_{DTI}: Binary interaction labels (0/1), y_{DTA}: Binding affinity values, Hyperparameters: Learning rate (η), batch size (B), epochs (E)

Ensure: Predicted interaction probabilities ($\hat{y}_{DTI}$), Predicted binding affinities ($\hat{y}_{DTA}$)

1: Step 1: Pre-processing

2: Tokenize drug SMILES into substructure → D_tokens

3: Tokenize protein sequence into amino acids → P_tokens

4: $D_tokens \rightarrow E_d \in R\hat{}(L_d \times d_e)$

5: $P_tokens \rightarrow E_p \in R\hat{}(L_p \times d_e)$

6: Step 2: Convolutional feature extraction

7: $F_d \leftarrow ReLU(Con1D_d(E_d))$

8: $F_p \leftarrow ReLU(Con1D_p(E_p))$

9: Step 3: Dual attention mechanism

10: $M_{c_d} = \sigma\big(MLP(GAP(F_d)\big) + MLP\big(GMP(F_d)\big)$

11: $F^{CA}_d = M_{c_d} \odot F_d$

12: $M_{c_p} = \sigma\left(MLP\big(GAP(F_p)\big)\right) + MLP\left(GMP\big(F_p\big)\right)$

13: $F^{CA}_p = M_{c_p} \odot F_p$

14: $M_{s_d} = \sigma(conv2D([AvgPool(F^{CA}_d); MaxPool(F^{CA}_d)]))$

15: $F^{SA}_d = M_{s_d} \odot F^{CA}_d$

16: $M_{s_p} = \sigma(conv2D([AvgPool(F^{CA}_p); MaxPool(F^{CA}_p)]))$

17: $F^{SA}_p = M_{s_p} \odot F^{CA}_p$

18: Step 4: Feature fusion and prediction

19: $z = [Flatten\big(F_d^{SA}\big); Flatten\big(F_p^{SA}\big)]$

20: $h = \sigma(W_h Z + b_h)$

21: $\hat{y}_{DTI} = Sigmoid(W_c h + b_c)$

22: $\hat{y}_{DTA} = W_r h + b_r$

23: Step 5: Compute loss and optimization

24: $\mathcal{L} = \lambda_1 * [-ylog(\hat{y}_{DTI}) - (1-y)log(1-\hat{y}_{DTI}) + \lambda_2 * (y_{DTI} - \hat{y}_{DTI})]$

25: Optimize with learning rate η

26: Step 6: for epoch = 1 *to* E:

27: **for** each batch (B):

28: Forward pass step 2 to 4

29: Compute loss (step 5)

30: Backpropagate and update weights

31: **end for**

32: **end for**

33: Step 7: return $\hat{y}_{DTI}$ and $\hat{y}_{DTA}$

Algorithm 1 Proposed nCNN-DA for DTI and DTA prediction.

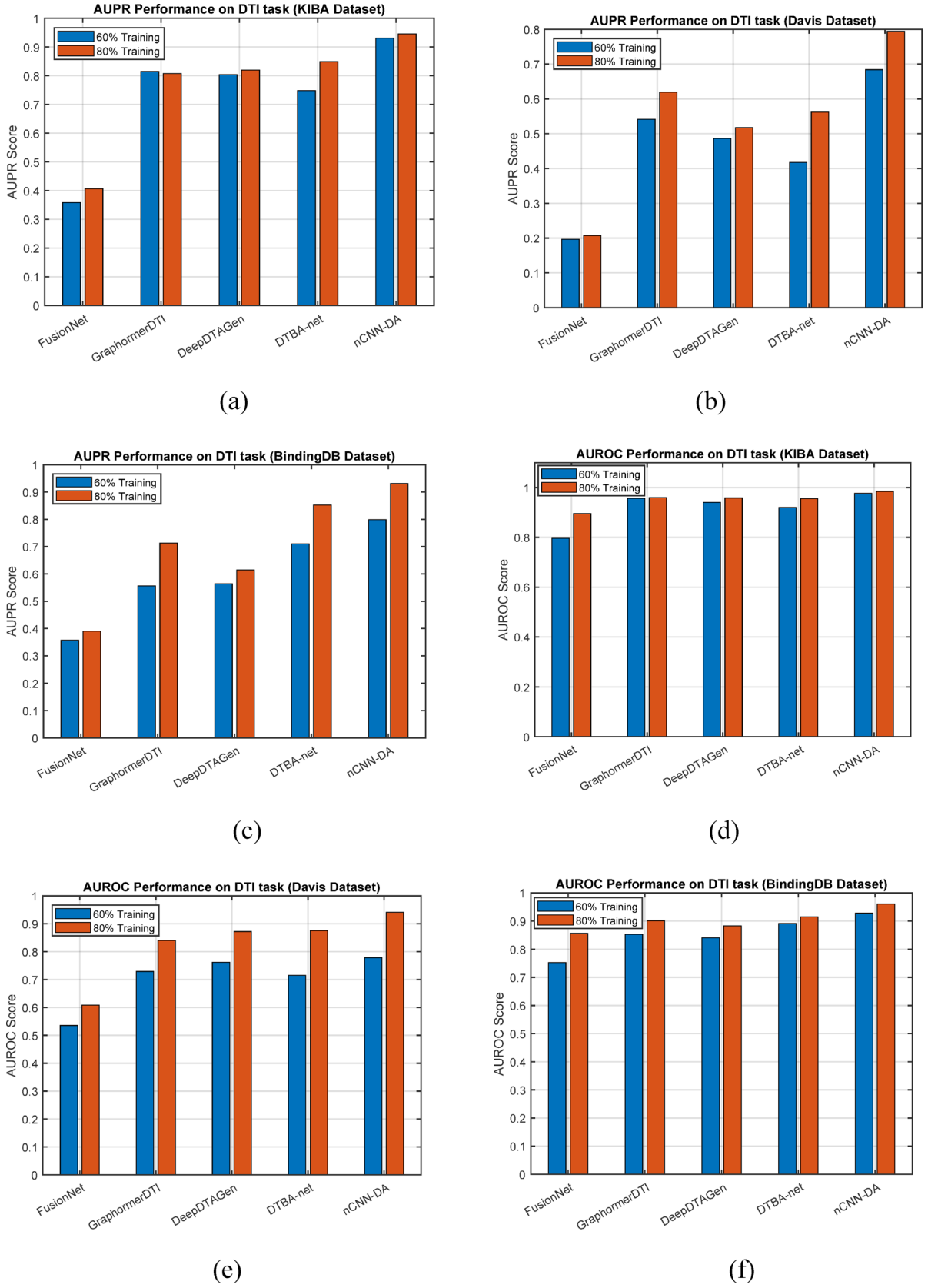


Fig. 2 AUPR and AUROC comparison on various models DTI task (KIBA, Davis, and BindingDB dataset)

Table 5 Ablation study of various model variant

Dataset	Model variant	AUPR	AUROC	MSE	Pearson correlation	Accuracy
KIBA	CNN only	0.8812	0.9624	0.1837	0.8615	0.941
	CNN + Channel Attention (CA)	0.9118	0.9715	0.1704	0.8759	0.961
	CNN + Spatial Attention (SA)	0.9036	0.9683	0.1739	0.8711	0.956
	CNN + CA + SA (nCNN-DA)	0.9457	0.9849	0.1559	0.8943	0.985
Davis	CNN only	0.7031	0.9012	0.3546	0.7024	0.914
	CNN + Channel Attention (CA)	0.7423	0.9218	0.3361	0.7215	0.932
	CNN + Spatial Attention (SA)	0.7315	0.9186	0.3427	0.7164	0.928
	CNN + CA + SA (nCNN-DA)	0.7944	0.9409	0.3189	0.739	0.955
BindingDB	CNN only	0.8723	0.9321	0.3284	0.7442	0.921
	CNN + Channel Attention (CA)	0.9027	0.9486	0.3105	0.7591	0.946
	CNN + Spatial Attention (SA)	0.8956	0.9443	0.3178	0.7538	0.941
	CNN + CA + SA (nCNN-DA)	0.9311	0.9609	0.2957	0.7733	0.975

Algorithm 1 presents the procedure of the proposed nCNN-DA model for predicting Drug–Target Interactions (DTIs) and Drug–Target Binding Affinities (DTAs). The drug and protein SMILES strings are tokenized, embedded, and normalized in the data pre-processing algorithm. The CNN feature extraction layer learns local substructure patterns in drug molecules and amino acid motifs in protein sequences with 1D convolution layers. A dual attention mechanism is adopted to improve the feature representation. The CA module emphasizes important biochemical feature channels, and the SA module emphasizes important spatial region of interaction between drug atoms and protein residues. The resultant drug and protein features are then concatenated and fed into fully connected layers to learn interaction features. The model is equipped with two prediction heads: a binary classification head (i.e. sigmoid activation) for the DTI prediction and a regression head (i.e. linear activation) for the DTA prediction (pKd). The model is trained with a hybrid loss function, which combined BCE and MSE, making it robust between classification and regression tasks.

Experimental setup and simulation parameters

The nCNN-DA model was developed in Python 3.10 with the TensorFlow 2.14 libraries. All experiments were performed on an Intel Xeon Silver 4310 CPU (2.1 GHz, 32 cores) and 128 GB DDR4 RAM. The drug molecules are represented only using SMILES sequences and Protein targets are represented using amino acid sequences in FASTA format. Each protein sequence is tokenized at the residue level (20 standard amino acids and padding token). Kd, Ki and IC50 affinities were normalised to pKd values and sequences were padded (or, if necessary, truncated) to a maximum length of 100 tokens for drug SMILES, and 1000 amino acids for proteins. We randomly divided the datasets (KIBA, Davis, and BindingDB) into 80% for training, 10% for validation, and 10% for testing based on stratified sampling while keeping balanced distribution of interactions.

The model was optimized by the algorithm Adam with an initial learning rate of 1e-4 (decayed with a factor of 0.95 in every 10 epochs), with a batch size of 64, and with a limit of 40 epochs and early stopping in case of absence of minimum loss in 10 consecutives epochs. The simulation parameters are shown in Table 4. For the hidden layers, we used ReLU as activation, Sigmoid for binary DTI prediction and Linear for continuous DTA prediction. The mixed loss function integrated BCE of DTI classification and MSE of DTA regression, and the full connection layers were added dropout (rate = 0.3). The model performance was assessed by 5-fold cross validation based on a series of different classification (Accuracy, AUROC, and AUPR) and regression (MSE, and Pearson correlation coefficient) metrics, to ensure the robust predictions and the good generalizability.

Performance evaluation and analysis

The assessment of nCNN-DA compared to the baseline methods such as DTBA-net [15], DeepDTAGen [16], GraphormerDTI [17], and FusionNet [18] on three benchmark datasets (such as KIBA, Davis, and BindingDB) demonstrates that our method outperforms the counterparts with significant advances in DTI classification, and DTA prediction tasks. The nCNN-DA achieves consistent improvement over the baseline models for all evaluation metrics and datasets. The gains owe to its two-tier attention mechanism, which can adaptively retrieve local substructure features and global interaction dependencies.

Figure 2 shows the comparisons on DTI task of nCNN-DA and other baseline models (FusionNet, GraphormerDTI, DeepDTAGen, DTBA-net), on using 3 benchmark DTI datasets (KIBA, Davis, BindingDB). The nCNN-DA

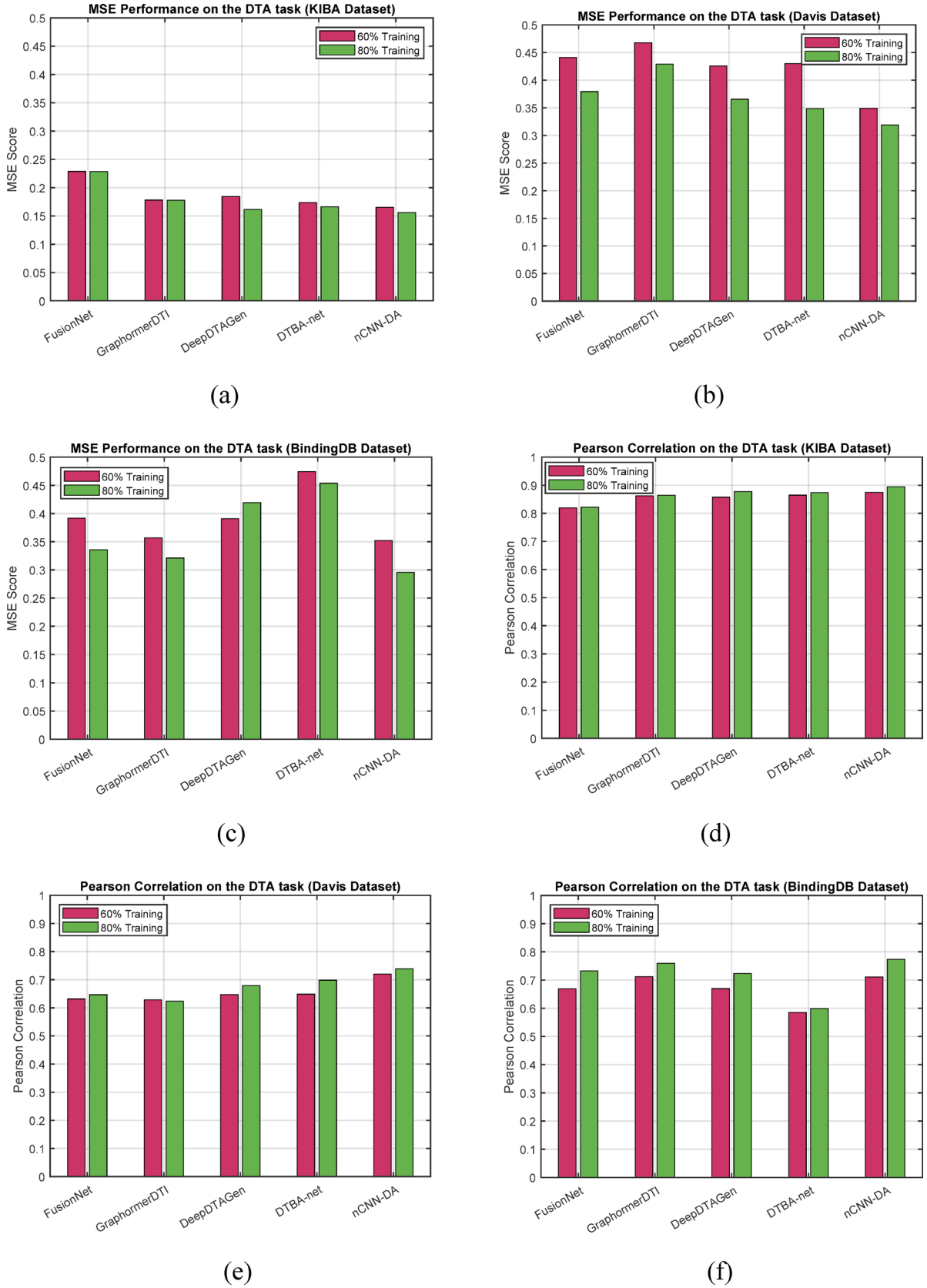
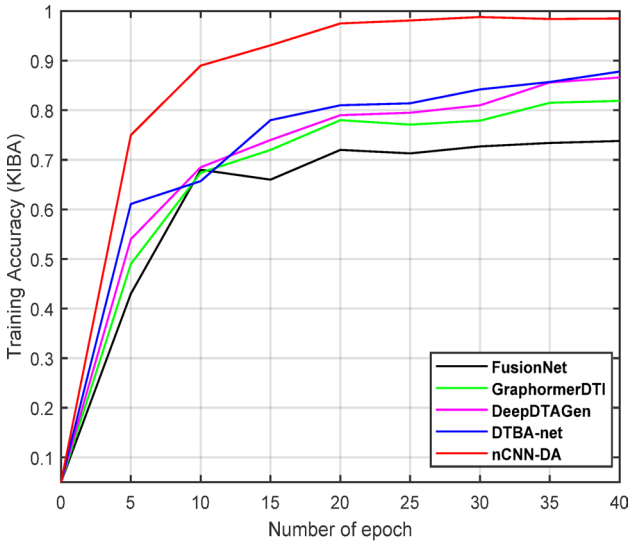
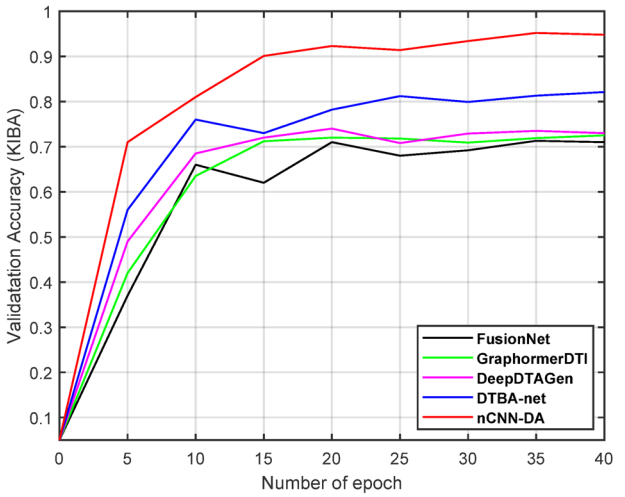


Fig. 3 MSE and Pearson Correlation comparison on various models DTA task (KIBA, Davis, and BindingDB datasets)

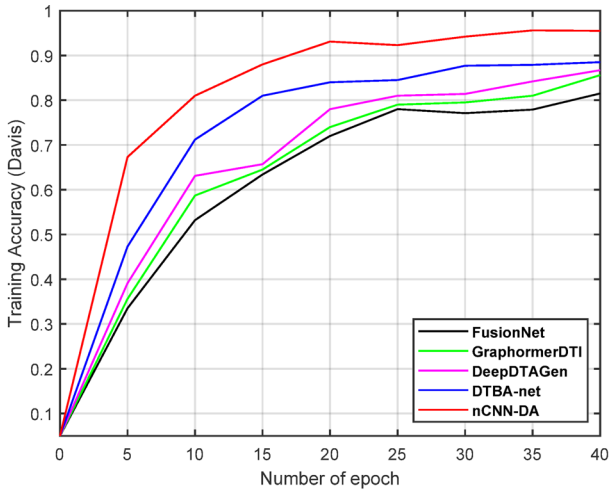


(a)

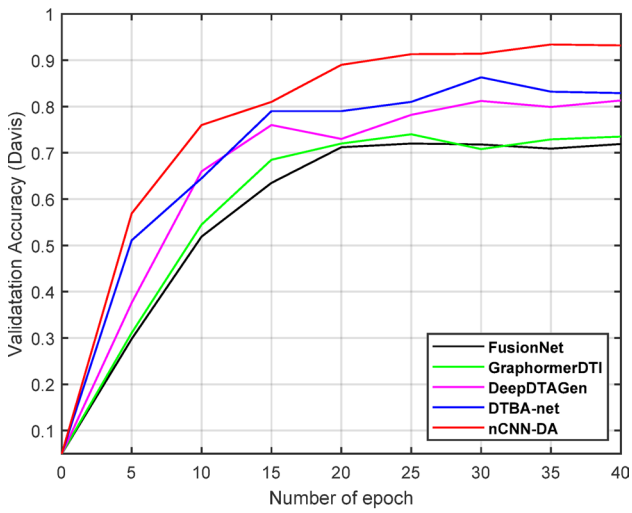


(b)

Fig. 4 Accuracy comparison on various models. **a** Training accuracy on KIBA dataset; **b** Validation accuracy on KIBA dataset



(a)



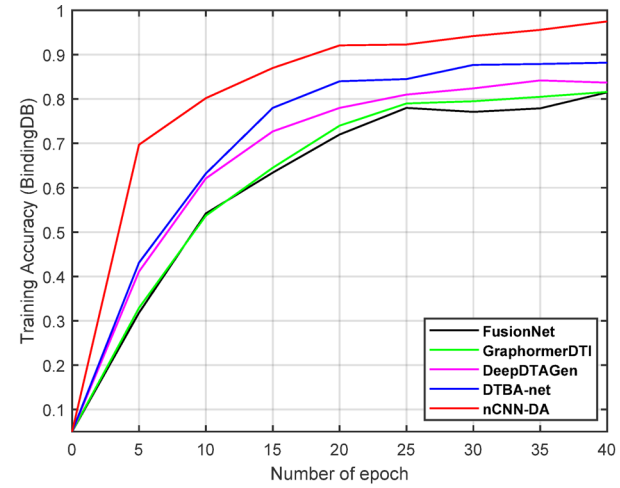
(b)

Fig. 5 Accuracy comparison on various models. **a** Training accuracy on Davis dataset; **b** Validation accuracy on Davis dataset

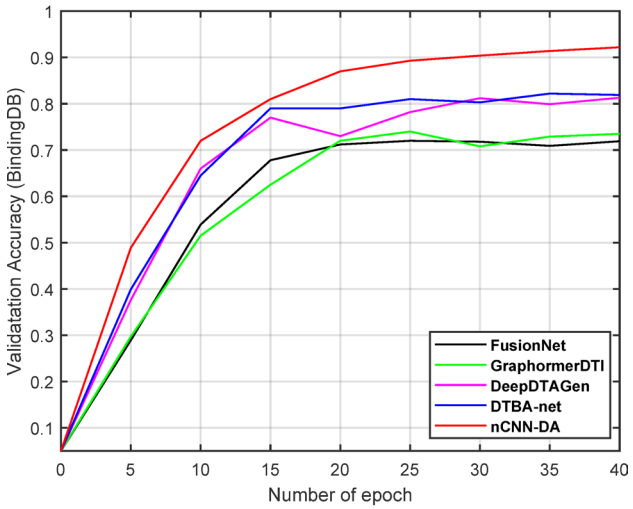
achieves the best performance over all baseline models on all the datasets, indicating that it is more powerful in discriminative ability between interacting and non-interacting drug-target pairs. Despite the challenge of using such a diverse dataset of drug and protein families of the BindingDB, nCNN-DA consistently has the highest AUPR and AUROC, which confirms its competitiveness in the context of large-scale and heterogeneous interaction prediction. The fact that the performance continues to increase demonstrates the effectiveness of the dual attention mechanism in encoding important biochemical and spatial interaction patterns, which are not sufficiently modeled by the baseline models. Table 5 shows the ablation study on various model variant with KIBA, Davis, and BindingDB datasets.

Figure 3 Comparison of MSE (Mean Squared Error) and Pearson Correlation results of nCNN-DA method comparing to the baseline models (FusionNet, GraphormerDTI, DeepDTAGen, DTBA-net) using KIBA, Davis and BindingDB databases. For panels (a–c), MSE performance is shown and for panels (d–f) Pearson Correlation performance is illustrated.

The performance of the proposed nCNN-DA, in terms of the lowest MSE, is always better than that of existing models for the prediction of binding affinity values, over all the datasets. On the KIBA dataset, nCNN-DA achieves an MSE of less than 0.16, which surpasses DTBA-net and



(a)



(b)

Fig. 6 Accuracy comparison on various models. **a** Training accuracy on BindingDB dataset; **b** Validation accuracy on BindingDB dataset

DeepDTAGen that have had competitive performance. The performance gains are more exaggerated in the Davis and BindingDB datasets, showing higher generalization of the nCNN-DA network that possesses the lowest predictive errors between nCNN.

In Pearson Correlation, nCNN-DA consistently has the highest correlation values, indicating that it has a better capturing capability to the linear fitting relationships between the predicted and real binding affinities. On the KIBA, the nCNN-DA obtains a correlation near 0.90, and on Davis and BindingDB, the the nCNN-DA even outperforms all the baseline models with correlations above 0.73 and 0.78, respectively.

Figure 4 compares the training and validation accuracy for the proposed nCNN-DA model and some baseline models (FusionNet, GraphormerDTI, DeepDTAGen and DTBA-net) on the KIBA dataset over 40 epochs. The training and validation accuracy is presented in panel (a) and (b), respectively. The proposed nCNN-DA model achieves the optimal accuracy in all the training steps, which close to 99%~100% in the training stage and over 95% in the validation stage, which demonstrates the rapid convergence and better generalization. On the contrary, baseline models have slower speed of convergence and get stuck with lower accuracy levels. DTBA-net and DeepDTAGen achieve relatively competitive performance but significantly worse convergence compared to nCNN-DA in later epochs. GraphormerDTI and FusionNet have a moderate performance with close to 80% validation accuracy and stabilizing performance suggests that they have a lower feature representation power.

Figure 5 shows the comparison of the training and validation accuracy of the proposed nCNN-DA model with those of baseline models (FusionNet, GraphormerDTI, DeepDTAGen, and DTBA-net) on the Davis dataset with 40 epochs. The training accuracy (a) and validation accuracy (b), respectively. The proposed nCNN-DA performs consistently better than all the baseline models and reaches the best accuracy rates in short convergence time. It is achieving over 95% training accuracy and over 90% validation accuracy, indicating good learning and not too bad generalization. DTBA-net has competitive performances against the baseline models, followed by DeepDTAGen, then GraphormerDTI and FusionNet are slower and the validation accuracy is lower compared to less than 80%.

Figure 6 shows the training and validation accuracy of the proposed nCNN-DA model compared with the baselines (FusionNet, GraphormerDTI, DeepDTAGen and DTBA-net) for 40 epochs on the BindingDB dataset. (a) shows training accuracy and (b) shows validation accuracy. The nCNN-DA is synergistic with all four sampling methods and also reaches the state-of-the-art performance, with a training accuracy of ~ 97% that keeps the validation accuracy over 90% for the diverse and heterogenic BindingDB (Table 5). In contrast, DTBA-net and DeepDTAGen have slightly lower training and validation accuracy, settled down but below 85%, while GraphormerDTI and FusionNet performed worse with validation accuracy at about 75% and 70%, respectively.

Figure 7 shows the Pearson Correlation performance results of nCNN-DA versus four baseline models (FusionNet, GraphormerDTI, DeepDTAGen, DTBA-net) with varying scale of labeled training data (20%, 40%, 60%, 80%) on three benchmark datasets (KIBA, Davis,

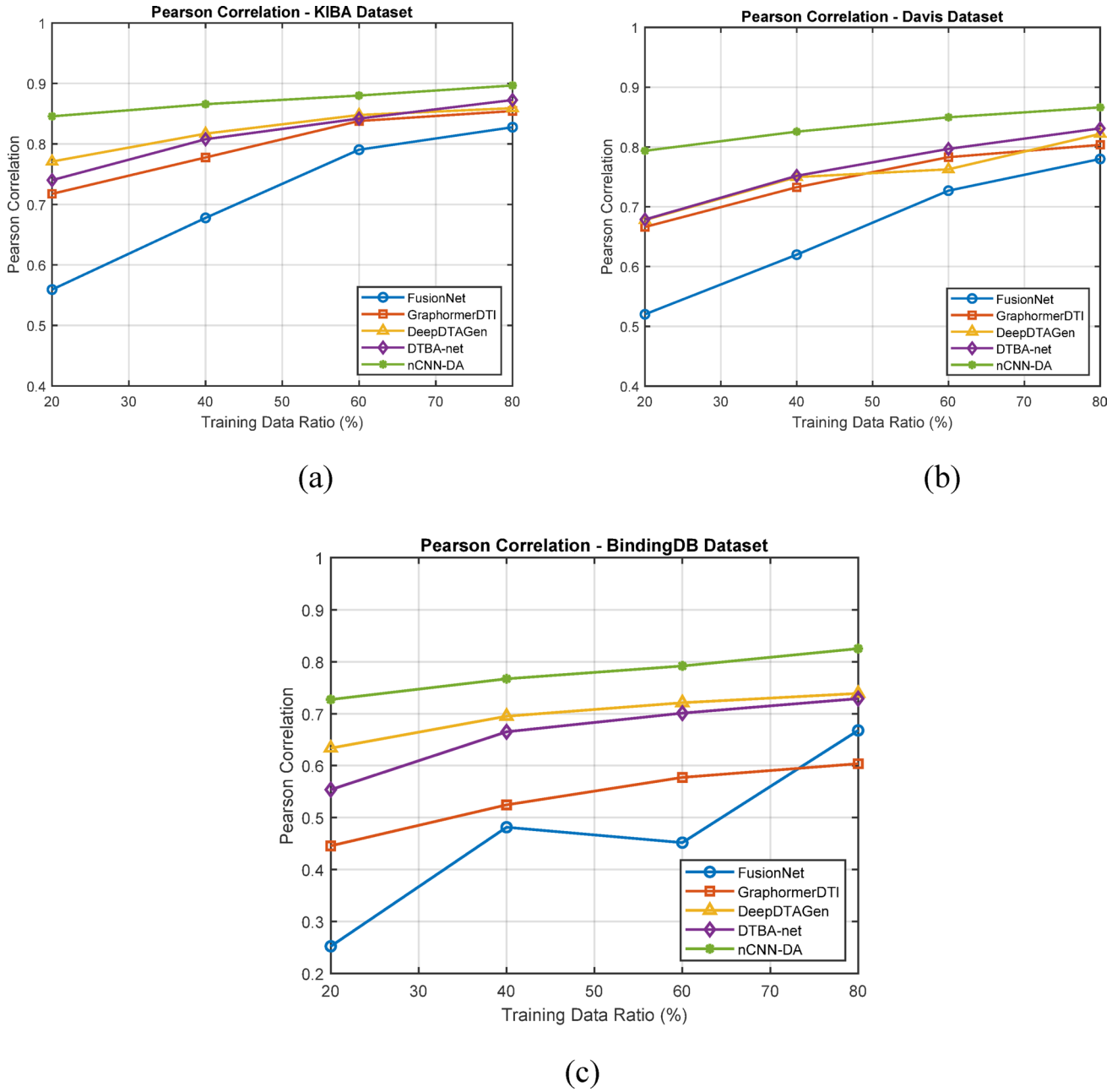


Fig. 7 Performance analysis of Pearson Correlation on the nCNN-DA and baseline model. **a** Different scale of label data on KIBA dataset, **b** Different scale of label data on Davis dataset, **c** Different scale of label data on BindingDB dataset

Table 6 Mean ±standard deviation across multiple runs

Model variant	Run1	Run2	Run3	Run4	Run5	Mean (%)	Mean ± Std
CNN only	0.939	0.942	0.944	0.941	0.943	94.18	94.18 ± 0.27
CNN + Channel Attention (CA)	0.958	0.962	0.964	0.96	0.961	96.1	96.1 ± 0.24
CNN + Spatial Attention (SA)	0.953	0.956	0.958	0.955	0.957	95.58	95.58 ± 0.16
CNN + CA + SA (nCNN-DA)	0.983	0.986	0.988	0.985	0.987	98.58	98.58 ± 0.11

Table 7 Computational complexity and efficiency analysis

Model	Parameters (Millions)	Training time/epoch (sec)	Total training time (min)	Inference time (ms)
DTBA-net	4.1	18	12	1.3
DeepDTAGen	6.8	27	18.5	2.1
FusionNet	9.5	42	29	3.8
GraphormerDTI	12.7	58	40.5	5.6
nCNN-DA (Proposed)	5.3	22	14.5	1.7

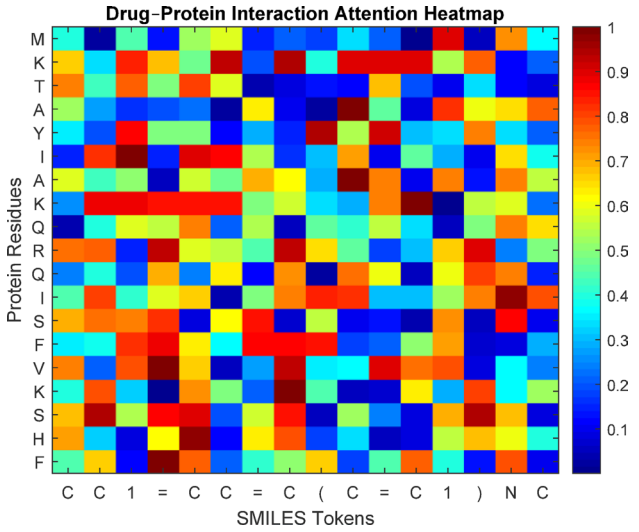


Fig. 8 Drug-protein spatial attention showing interaction importance between SMILE tokens and protein residue

BindingDB). Across all the three datasets, nCNN-DA provides the best Pearson Correlation values, demonstrating that it can better capture the real linear relationship between experimentally measured binding affinities and predicted values. On the KIBA dataset (Fig. 7a), comparable to that at 80% as DTBA-net and DeepDTAGen reach approximately 0.87 correlation; while DTBA-net and DeepDTAGen peak below 0.88. Likewise, for the Davis dataset (Fig. 7b), nCNN-DA consistently performs better with increasing training ratio up to around 0.87 correlation, while DTBA-net and DeepDTAGen are stable at 0.82–0.83. On the BindingDB dataset (Fig. 7c), but with more molecular variety nCNN-DA is still the best, with around 0.78 correlation, obviously better than GraphormerDTI and FusionNet, whose correlation is less than 0.70. Table 6 shows the mean ±standard deviation across multiple runs on various model variants.

Table 7 shows the computational complexity and efficiency analysis, which report the parameter count, training time per epoch, total training time, and inference time for nCNN-DA and all baseline models.

Drug-protein interaction heatmap analysis

To inspect the proposed nCNN-DA model is interpreted, as shown in Fig. 8, analyzed the spatial attention weights calculated by the dual attention mechanism using a drug–protein attention heatmap. The heatmap shows interaction importance between drug SMILES tokens and protein amino acid residues. In each block, we see the model's attention weight for that atom-residue interaction. Warmer colors (red and orange) are associated with higher attention scores, meaning that these regions were more informative for determining the interaction relevance of the underlying complex binding; cooler colors (blue and green), then, represent lower contribution to the estimated overall binding affinity.

Table 8 Performance evaluation metrics analysis of nCNN-DA and baseline model

Benchmark datasets	Models	AUPR	AUROC	MSE	Person correlation	Accuracy
KIBA	FusionNet	0.4065	0.895289	0.228728	0.822367	0.738
	GraphormerDTI	0.8072	0.96013	0.177841	0.864202	0.819
	DeepDTAGen	0.8198	0.958303	0.1615	0.877436	0.866
	DTBA-net	0.8487	0.956058	0.166083	0.873194	0.878
	nCNN-DA	0.9457	0.984983	0.155908	0.894383	0.985
Davis	FusionNet	0.2069	0.608274	0.379317	0.646606	0.817
	GraphormerDTI	0.6194	0.840181	0.429144	0.624075	0.856
	DeepDTAGen	0.5176	0.872193	0.365466	0.678609	0.867
	DTBA-net	0.5622	0.875577	0.348678	0.698241	0.885
	nCNN-DA	0.7944	0.940913	0.318976	0.739032	0.955
BindingDB	FusionNet	0.3902	0.856086	0.335613	0.732753	0.815
	GraphormerDTI	0.7133	0.901478	0.321079	0.759216	0.816
	DeepDTAGen	0.6148	0.882834	0.419345	0.723043	0.837
	DTBA-net	0.8526	0.915217	0.45371	0.598734	0.882
	nCNN-DA	0.9311	0.960923	0.295689	0.77327	0.975

As observed in Fig. 8, attention is not evenly distributed over the sequence space and the model focuses higher attention weights on a subset of SMILES tokens and protein residues. In the chosen example, many SMILES tokens representing chemically relevant molecular fragments (aromatic C atoms + functional groups ("C = C" and heteroatom-connected structures) trigger strong attention response. At the same time a sub-set of protein residues such as lysine (K), arginine (R), serine (S) and histidine (H) emerge higher attention scores, indicating that these residue plays an important role in interaction mechanism.

Result discussion

Experimental comparison against four state-of-the-art benchmark models (FusionNet, GraphormerDTI, DeepDTAGen, and DTBA-net) using three benchmark datasets (KIBA, Davis, and BindingDB) evidence the superior prediction capability of our nCNN-DA in both DTI and DTA prediction. Overall, on all the datasets nCNN-DA performs better because of its bidirectional attention mechanism focusing on highlighting both important feature channels (Channel Attention) and significant spatial interaction regions (Spatial Attention) concurrently. The combination of convolutional feature extraction with attention makes the feature representation rich and the separation of the important drug–protein interaction sites is more significant. But for these baseline models, they perform well in AUROC and Pearson correlation but poor in AUPR and MSE, which demonstrates their limitations in precise binding affinity prediction.

In Table 8, The AUPR is 0.9457 and AUROC is 0.9849, which measure its capacity to separate interacting pairs from non-interacting ones are superior to all other nCNN-DA models. In addition, the lowest MSE (0.1559) means it could better predict binding affinities than the base models. In terms of correlation, nCNN-DA obtains the highest Pearson correlation (0.8943), indicating the strong strength to grasp the linear relationship between predicted and true binding affinity scores. The highest accuracy (98.5%) additionally validates its robust DTI classification.

On Davis dataset, nCNN-DA outperforms all the competitor models again, reaching the best AUPR (0.7944), AUROC (0.9409) and accuracy (95.5%). Moreover, its MSE (0.3189) is less than the MSE of all the other models which means that it is better predictor for binding affinity. The Pearson correlation value of nCNN-DA (0.7390) is higher than the other models, although DTBA-net and DeepDTAGen also have competitive correlations. Overall, the performance gain on Davis illustrates nCNN-DA's generalizability, especially for kinase inhibitors where attention-based feature refinement is more advantageous, as it aids in discovering informative drug–protein interactions. On the BindingDB dataset, the efficacy of nCNN-DA is again verified, and receives the best AUPR (0.9311), AUROC (0.9609) and accuracy (97.5%). The model also achieves the lowest MSE (0.2957), indicating its superior regression performance for a wider-variety of drug–target pairs other than the kinase-specific interaction. Using Pearson correlation (0.7733) as comparison, this performance of nCNN-DA is marginally lower than GraphormerDTI (0.7592), but significantly improved than DeepDTAGen (0.7230), demonstrating its excellent performance across multiple molecular scaffolds and protein families.

Conclusion

In this work, we developed nCNN-DA, a convolutional neural network based dual attention mechanism, for precise DTI and DTA prediction. The advantages of incorporating channel and spatial attention, as well as CNN, for precisely capturing both local biochemical characteristics and global structural dependencies in modeling drug-protein interactions are explicitly demonstrated.

Extensive experimental results on three benchmark datasets, namely, kiba, davis, and bindingdb are presented, which show that nCNN-DA outperforms the state-of-the-art models such as FusionNet, GraphormerDTI, DeepDTAGen, and DTBA-net. For the KIBA dataset, the best performance was observed for nCNN-DA, which achieved an accuracy of 98.5% and a minimum MSE of 0.1559, compared to DTBA-net (with respective accuracies and MSEs of 87.8% and 0.1660). Its performance achieved an accuracy of 95.5%, with an MSE of 0.3189 on the Davis dataset, outperforming DTBA-net (88.5% accuracy, 0.3486 MSE). In a similar trend, nCNN-DA achieved an accuracy of 97.5% and an MSE of 0.2957 on the BindingDB dataset, outperforming all baseline models. Moreover, nCNN-DA consistently obtained higher Pearson Correlation (up to 0.8943) and AUPR/AUROC scores on all datasets, demonstrating its ability to capture informative interaction patterns.

The experiments demonstrate that nCNN-DA not only provides excellent DTI prediction accuracy but also yields a reliable prediction of binding affinities in DTA prediction, serving as a robust and general framework for drug discovery applications. The high accuracy and low MSE over various datasets confirm its promising applications in drug repositioning and virtual screening. In future framings, we will consider incorporating multi-modal features, such as 3D structural information and graph-based

molecular representations, to improve prediction accuracy and interpretability.

Author contributions All authors have equal contribution.

Funding The authors declare that they have no competing financial interests in this paper.

Data availability Datasets are available publicly to use in this work on https://www.kaggle.com/datasets/rajaryan2315/davis-and-kiba-datasets and https://doi.org/10.1038/nbt.1990 and https://string-db.org/.

Declarations

Conflict of interest The authors declare no competing interests.

Informed consent All the authors have given their consent for the publication of this article.

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